Time Series — Solutions

Practice Exam — Week 6: Fourier Theory

EPFL, MATH-342 · Prof. S. C. Olhede

Problem 1 (5 points)

Statement.

Throughout this problem $g \in L^1$ has continuous-time Fourier transform

$$G(f) = \int_{-\infty}^{\infty} g(t)\, e^{-2\pi i f t}\, dt,$$

and for a derived function h we write H for its transform. We use the four elementary Fourier-transform properties (conjugation, scaling, shift, linearity).

(a) *Prove the* conjugation *and* time-shift *rules: if $h(t) = \overline{g(t)}$ then $H(f) = \overline{G(-f)}$, and if $h(t) = g(t+\tau)$ then $H(f) = G(f)\, e^{2\pi i f \tau}$.* *(2 pts)*

(b) *Combining scaling and shifting, show that the* modulated *signal $h(t) = g(\alpha t)\, e^{2\pi i f_0 t}$ (with $\alpha \neq 0$, $f_0 \in \mathbb{R}$ fixed) has transform*

$$H(f) = \frac{1}{|\alpha|}\, G\!\left(\frac{f - f_0}{\alpha}\right).$$

State in one sentence what each of the two operations (scaling by α, modulation by $e^{2\pi i f_0 t}$) does to the spectrum. *(2 pts)*

(c) *Use part (a) to show that if g is* real and even *(so $g(t) = \overline{g(t)} = g(-t)$), then its transform G is real and even.* *(1 pt)*

Solution.

(a) *Conjugation.* Pull the conjugate outside the integral; it flips the sign of the exponent:

$$H(f) = \int_{-\infty}^{\infty} \overline{g(t)}\, e^{-2\pi i t f}\, dt = \overline{\int_{-\infty}^{\infty} g(t)\, e^{2\pi i t f}\, dt} = \overline{\int_{-\infty}^{\infty} g(t)\, e^{-2\pi i t(-f)}\, dt} = \overline{G(-f)}.$$

Time shift. Insert $1 = e^{-2\pi i f \tau} e^{2\pi i f \tau}$ to complete the exponent:

$$H(f) = \int_{-\infty}^{\infty} g(t+\tau)\, e^{-2\pi i f t}\, dt = \int_{-\infty}^{\infty} g(t+\tau)\, e^{-2\pi i f (t+\tau)}\, dt\; e^{2\pi i f \tau} = G(f)\, e^{2\pi i f \tau},$$

where in the last step we substituted $s = t + \tau$ ($ds = dt$, limits unchanged) so the integral is exactly $G(f)$.

(b) Build h in two steps. First the *scaling* rule: with $p(t) = g(\alpha t)$, substitute $s = \alpha t$ (so $dt = ds/|\alpha|$, the $|\alpha|$ arising from reversing the limits when $\alpha < 0$),

$$P(f) = \int_{-\infty}^{\infty} g(\alpha t)\, e^{-2\pi i f t}\, dt = \frac{1}{|\alpha|} \int_{-\infty}^{\infty} g(s)\, e^{-2\pi i (f/\alpha) s}\, ds = \frac{1}{|\alpha|}\, G\!\left(\frac{f}{\alpha}\right).$$

Now $h(t) = p(t)\, e^{2\pi i f_0 t}$ is a pure *modulation*; modulation in time is a frequency shift, since

$$H(f) = \int_{-\infty}^{\infty} p(t)\, e^{2\pi i f_0 t}\, e^{-2\pi i f t}\, dt = \int_{-\infty}^{\infty} p(t)\, e^{-2\pi i (f - f_0) t}\, dt = P(f - f_0).$$

Substituting P,

$$\boxed{H(f) = P(f - f_0) = \frac{1}{|\alpha|}\, G\!\left(\frac{f - f_0}{\alpha}\right).}$$

In words: scaling time by α *inversely* stretches the frequency axis (by $1/\alpha$) and rescales the height by $1/|\alpha|$; modulating by $e^{2\pi i f_0 t}$ *translates* the whole spectrum to be centred at f_0.

(c) Let g be real and even. *Real* ($\overline{g(t)} = g(t)$) means g equals its own conjugate, so by the conjugation rule of (a),

$$G(f) = \overline{G(-f)} \iff G(-f) = \overline{G(f)} \qquad \text{(Hermitian symmetry)}.$$

Even ($g(-t) = g(t)$) is the scaling rule of (b) with $\alpha = -1$ (so $|\alpha| = 1$), giving $G(-f) = G(f)$ (G even). Combining the two,

$$G(f) = G(-f) = \overline{G(f)},$$

so $G(f)$ equals its own conjugate, i.e. G is **real**; and $G(f) = G(-f)$ says G is **even**. (This is exactly why a valid SDF, the transform of the real even ACVS, is real and even.)

Problem 2 (5 points)

Statement.

Recall the continuous-time convolution of $g, h \in L^1$,

$$(g * h)(t) = \int_{-\infty}^{\infty} g(t-y)\, h(y)\, dy,$$

and that upper-case letters denote Fourier transforms.

(a) *Prove the* convolution theorem*: if $u = g * h$ then $U = G \cdot H$. State explicitly the Fubini step and why the hypotheses $g, h \in L^1$ justify it.* *(3 pts)*

(b) (Filtering.) *A signal $x \in L^1$ is passed through a linear filter with impulse response $a \in L^1$, producing $y = a * x$. Using part (a), give Y in terms of X and the transfer function A. If the filter is the running-average kernel $a(t) = \frac{1}{2T}\,\mathbf{1}_{\{|t|\le T\}}$, compute $A(f)$ in closed form and verify $A(0) = 1$.* *(2 pts)*

Solution.

(a) By definition of the transform and of convolution,

$$\begin{aligned}
U(f) &= \int_{-\infty}^{\infty} u(x)\, e^{-2\pi i x f}\, dx = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x-y)\, h(y)\, dy \right) e^{-2\pi i x f}\, dx \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x-y)\, e^{-2\pi i x f}\, dx \right) h(y)\, dy \qquad \text{(Fubini: swap the order of integration)} \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(s)\, e^{-2\pi i s f}\, ds \right) e^{-2\pi i y f}\, h(y)\, dy \qquad (s = x - y,\ dx = ds) \\
&= G(f) \int_{-\infty}^{\infty} h(y)\, e^{-2\pi i y f}\, dy = G(f)\, H(f).
\end{aligned}$$

The Fubini interchange is legitimate because the double integral is absolutely convergent:

$$\int_{-\infty}^{\infty}\!\int_{-\infty}^{\infty} |g(x-y)\, h(y)|\, dx\, dy = \int_{-\infty}^{\infty} |g(s)|\, ds \int_{-\infty}^{\infty} |h(y)|\, dy = \|g\|_1\, \|h\|_1 < \infty,$$

using the change of variables $s = x - y$ and the hypotheses $g, h \in L^1$. Hence $U = G \cdot H$: the transform of a convolution is the product of the transforms.

(b) With $y = a * x$, part (a) gives immediately

$$Y(f) = A(f)\, X(f),$$

where $A(f) = \int_{-\infty}^{\infty} a(t)e^{-2\pi ift}\,dt$ is the *transfer function*: filtering multiplies the spectrum frequency-by-frequency by A. For the running average $a(t) = \frac{1}{2T}\mathbf{1}_{\{|t|\le T\}}$,

$$A(f) = \frac{1}{2T}\int_{-T}^{T} e^{-2\pi ift}\,dt = \frac{1}{2T}\cdot\frac{e^{-2\pi ift}}{-2\pi if}\Bigg|_{t=-T}^{t=T} = \frac{1}{2T}\cdot\frac{e^{2\pi ifT}-e^{-2\pi ifT}}{2\pi if} = \frac{\sin(2\pi fT)}{2\pi fT}.$$

At $f=0$ the indeterminate form has limit 1 (since $\sin x/x \to 1$); equivalently $A(0) = \frac{1}{2T}\int_{-T}^{T} 1\,dt = 1$. So the filter preserves the zero-frequency (mean/DC) component and attenuates higher frequencies — a low-pass filter, as a running average should be.

Problem 3 (5 points)

Statement.

This problem concerns the finite-data Fourier transform, the Dirichlet kernel, and aliasing. Take sampling interval $\Delta = 1$ and observation times $t = 0, 1, \ldots, n-1$.

(a) *The transform of the finite rectangular window is the* ***Dirichlet kernel***

$$D_n(f) = \sum_{t=0}^{n-1} e^{-2\pi ift}.$$

Sum this finite geometric series and show that

$$D_n(f) = \frac{\sin(n\pi f)}{\sin(\pi f)}\, e^{-\pi i(n-1)f}, \qquad \text{so} \qquad |D_n(f)| = \left|\frac{\sin(n\pi f)}{\sin(\pi f)}\right|.$$

(2 pts)

(b) *Show that $|D_n(f)|$ vanishes exactly at the nonzero* ***Fourier frequencies*** *$f_k = k/n$, $k = 1, \ldots, n-1$, while $D_n(0) = n$. (This is why the periodogram of a pure sinusoid at a Fourier frequency does not leak onto the other Fourier frequencies.)* *(1.5 pts)*

(c) (Aliasing / Nyquist.) *Let $\{X_t\}$ be obtained by sampling a continuous-time process at interval Δ, and recall the aliasing identity for the sampled spectral density,*

$$\mathcal{S}(f) = \sum_{k\in\mathbb{Z}} \mathcal{S}_{\mathrm{c}}\left(f + \frac{k}{\Delta}\right), \qquad |f| \le \frac{1}{2\Delta}.$$

Suppose the continuous SDF is band-limited triangular, $\mathcal{S}_{\mathrm{c}}(f) = (1-|f|)\,\mathbf{1}_{\{|f|\le 1\}}$. For $\Delta = 1$ (Nyquist frequency $\frac{1}{2}$) compute the aliased SDF $\mathcal{S}(f)$ on $|f| \le \frac{1}{2}$, and state whether aliasing has occurred and why. (1.5 pts)

Solution.

(a) The sum is a finite geometric series with ratio $r = e^{-2\pi if}$. For $f \notin \mathbb{Z}$ (so $r \ne 1$),

$$D_n(f) = \sum_{t=0}^{n-1} r^t = \frac{1-r^n}{1-r} = \frac{1-e^{-2\pi inf}}{1-e^{-2\pi if}}.$$

Factor a half-angle phase out of numerator and denominator:

$$\begin{aligned} D_n(f) &= \frac{e^{-\pi inf}\left(e^{\pi inf} - e^{-\pi inf}\right)}{e^{-\pi if}\left(e^{\pi if} - e^{-\pi if}\right)} = \frac{e^{-\pi inf}\,(2i)\sin(n\pi f)}{e^{-\pi if}\,(2i)\sin(\pi f)} \\ &= \frac{\sin(n\pi f)}{\sin(\pi f)}\, e^{-\pi i(n-1)f}, \end{aligned}$$

using $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$. The phase factor has modulus 1, so

$$|D_n(f)| = \left|\frac{\sin(n\pi f)}{\sin(\pi f)}\right|.$$

(b) The numerator $\sin(n\pi f)$ vanishes exactly when $nf \in \mathbb{Z}$, i.e. at $f = k/n$, $k \in \mathbb{Z}$. The denominator $\sin(\pi f)$ vanishes only when $f \in \mathbb{Z}$. At the Fourier frequencies $f_k = k/n$ for $k = 1, \ldots, n-1$ we have $f_k \notin \mathbb{Z}$ (since $0 < k/n < 1$), so the denominator is nonzero while the numerator is zero:

$$|D_n(f_k)| = \frac{|\sin(k\pi)|}{|\sin(\pi k/n)|} = \frac{0}{\sin(\pi k/n)} = 0, \qquad k = 1, \ldots, n-1.$$

At $f = 0$ both vanish; resolving the limit (or summing directly $\sum_{t=0}^{n-1} 1$),

$$D_n(0) = \lim_{f\to 0} \frac{\sin(n\pi f)}{\sin(\pi f)} = \lim_{f\to 0} \frac{n\pi f}{\pi f} = n.$$

So $|D_n|$ has a peak of height n at the origin and is exactly zero at every *other* Fourier frequency. Consequently the finite transform of a sinusoid sitting at one Fourier frequency contributes nothing at the remaining Fourier frequencies: there is no spectral leakage *between* Fourier frequencies for an on-grid sinusoid.

(c) With $\Delta = 1$ the Nyquist frequency is $\frac{1}{2}$ and the aliasing identity reads $\mathcal{S}(f) = \sum_{k\in\mathbb{Z}} \mathcal{S}_{\mathrm{c}}(f+k)$ on $|f| \le \frac{1}{2}$. Since $\mathcal{S}_{\mathrm{c}}$ is supported on $[-1, 1]$, for $|f| \le \frac{1}{2}$ only $k \in \{-1, 0, 1\}$ can contribute:

$$\mathcal{S}(f) = \mathcal{S}_{\mathrm{c}}(f-1) + \mathcal{S}_{\mathrm{c}}(f) + \mathcal{S}_{\mathrm{c}}(f+1).$$

Evaluate each term on $|f| \le \frac{1}{2}$:

- $\mathcal{S}_{\mathrm{c}}(f) = 1 - |f|$ (always in support);
- $\mathcal{S}_{\mathrm{c}}(f+1) = 1 - |f+1|$ is nonzero only when $|f+1| \le 1$, i.e. $f \le 0$, where it equals $1-(1+f) = -f$;
- $\mathcal{S}_{\mathrm{c}}(f-1) = 1 - |f-1|$ is nonzero only when $|f-1| \le 1$, i.e. $f \ge 0$, where it equals $1-(1-f) = f$.

Hence

$$\mathcal{S}(f) = \begin{cases} (1-|f|) + (-f) = (1+f) + (-f) = 1, & -\frac{1}{2} \le f \le 0, \\ (1-|f|) + f = (1-f) + f = 1, & 0 \le f \le \frac{1}{2}, \end{cases}$$

that is, $\boxed{\mathcal{S}(f) \equiv 1 \ \text{ on } \ |f| \le \tfrac{1}{2}.}$ **Aliasing has occurred.** The continuous spectrum had support out to $|f| = 1$, i.e. *beyond* the Nyquist frequency $\frac{1}{2}$; sampling at $\Delta = 1$ folds that excess high-frequency power back into the band $[-\frac{1}{2}, \frac{1}{2}]$. Here the folded triangular tails fill in exactly the missing area, turning a triangular continuous SDF into a *flat* (white-looking) sampled SDF — the high-frequency content has been irretrievably mixed into the low frequencies.

Problem 4 (5 points)

Statement.

(Revision of Week 5: differencing & SARIMA.) *Consider the signal-plus-noise model $X_t = \mu_t + Y_t$, where μ_t is a deterministic component and $\{Y_t\}$ is a zero-mean stationary process with autocovariance sequence γ_τ. Recall $\nabla = I - B$, $\nabla_{(s)} = I - B^s$.*

(a) *Let $\mu_t = a + bt + s_t$ with a* monthly *season $s_{t+12} = s_t$. Compute $\nabla\nabla_{(12)} X_t$ explicitly as a linear combination of the Y's, verify its mean is 0, and explain in one sentence why it is weakly stationary. (2 pts)*

(b) *Now let $\{X_t\}$ be the multiplicative seasonal process* $\mathrm{SARMA}(0,1) \times (1,0)_{12}$,

$$(1 - \Phi B^{12}) X_t = (1 - \theta B)\varepsilon_t, \qquad |\Phi| < 1.$$

By inverting the seasonal AR factor, obtain the $\mathrm{MA}(\infty)$ *weights ψ_j in $X_t = \sum_{j\ge 0} \psi_j \varepsilon_{t-j}$ (closed form for ψ_{12l} and ψ_{12l+1}; all other $\psi_j = 0$). (2 pts)* *(c) Using $\gamma_\tau = \sigma^2 \sum_{j\ge 0} \psi_j \psi_{j+\tau}$, compute γ_0 and γ_1 in closed form, and evaluate $\rho_1 = \gamma_1/\gamma_0$ for $\theta = \frac{1}{2}$. (1 pt)*

Solution.

(a) Apply the seasonal difference first. Using $s_t = s_{t-12}$,

$$\nabla_{(12)} X_t = X_t - X_{t-12} = \big(a + bt + s_t + Y_t\big) - \big(a + b(t-12) + s_{t-12} + Y_{t-12}\big) = 12b + \big(Y_t - Y_{t-12}\big).$$

The period-12 season is annihilated and the linear trend collapses to the constant $12b$. Now apply $\nabla = I - B$, which kills the constant:

$$\nabla\,\nabla_{(12)} X_t = \big(12b + Y_t - Y_{t-12}\big) - \big(12b + Y_{t-1} - Y_{t-13}\big) = \boxed{Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}}.$$

This is a *fixed* finite linear combination of $\{Y_t\}$, so its mean is $\mathbb{E}[Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}] = 0$, and being a fixed linear filter of a stationary process its autocovariance depends on the lag τ only (no t); hence it is weakly stationary. All t-dependence (trend and season) has been removed.

(b) Solve the model for X_t: $X_t = \Phi X_{t-12} + \varepsilon_t - \theta\varepsilon_{t-1}$, a seasonal AR(1) at lag 12 driven by an MA(1) input. Invert the seasonal factor as a geometric series in B^{12} (valid since $|\Phi| < 1$),

$$\frac{1}{1 - \Phi B^{12}} = \sum_{l\ge 0} \Phi^l B^{12l},$$

so that

$$X_t = \sum_{l\ge 0} \Phi^l B^{12l} \big(\varepsilon_t - \theta\varepsilon_{t-1}\big) = \sum_{l\ge 0} \Phi^l \big(\varepsilon_{t-12l} - \theta\, \varepsilon_{t-12l-1}\big).$$

Reading off the coefficient of ε_{t-j},

$$\boxed{\psi_{12l} = \Phi^l, \qquad \psi_{12l+1} = -\theta\,\Phi^l \quad (l \ge 0), \qquad \psi_j = 0 \text{ otherwise.}}$$

(Check: $\psi_0 = 1,\ \psi_1 = -\theta,\ \psi_{12} = \Phi,\ \psi_{13} = -\theta\Phi, \dots$)

(c) Use $\gamma_\tau = \sigma^2 \sum_{j\ge 0} \psi_j \psi_{j+\tau}$. The only nonzero taps are at $j \equiv 0$ and $j \equiv 1 \pmod{12}$.

Lag 0*:*

$$\gamma_0 = \sigma^2 \sum_{l\ge 0} \big(\psi_{12l}^2 + \psi_{12l+1}^2\big) = \sigma^2 \sum_{l\ge 0} \big(\Phi^{2l} + \theta^2\Phi^{2l}\big) = \frac{(1+\theta^2)\,\sigma^2}{1 - \Phi^2}.$$

Lag 1*:* the only surviving pairs are $\psi_{12l}\,\psi_{12l+1} = \Phi^l(-\theta\Phi^l) = -\theta\Phi^{2l}$, so

$$\gamma_1 = \sigma^2 \sum_{l\ge 0} \big(-\theta\Phi^{2l}\big) = \frac{-\theta\,\sigma^2}{1 - \Phi^2}.$$

Therefore

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\theta}{1 + \theta^2}.$$

For $\theta = \frac{1}{2}$: $\rho_1 = \dfrac{-1/2}{1 + 1/4} = \dfrac{-1/2}{5/4} = -\dfrac{2}{5} = -0.4$. (The autocorrelation is governed entirely by the MA(1) side-factor near the origin; the seasonal AR(1) produces analogous spikes at the seasonal lags $12, 24, \dots$ of relative size Φ^k.)